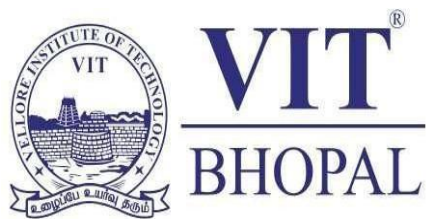


VITYARTHI – PROJECT



Project title : Online Quiz Management System

Course title : Programming in Java

Course code : CSE 2006

Course type : Flipped course

Course credits : 3

Professor : Dr. Rizwan Ur Rahman

Slot : F11 + F12

Submitted by : Gokul Prasad. K

Registration number : 25BAI10148

Submission date : 17/09/2026

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INTRODUCTION :

QuizVerse is a Java-based system that is used to manage quizzes, and which allows users to conduct quizzes in a simplistic and interactive way. The system enables users to register and log in, answer multiple choice questions, and get scores on their performance. There is also an admin part of the system, as well as the leader board where scores of the students are shown.

The Java and Object-Oriented Programming concepts used in the project include but are not limited to classes, inheritance, abstraction, polymorphism, exception handling, ArrayList and multithreading. A timer is also included for each question and shuffling of the questions is done to ensure that the quiz becomes more interactive. This helps in demonstrating the applicability of core Java concepts in a practical way.

The major aim of the QuizVerse project was to come up with a quiz application that is simplistic in its usage, while at the same time applying the Java programming concepts that have been learnt. Improvements to the project could be made through including the database, graphical user interface, quiz categories among others.

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PROBLEM STATEMENT :

Usually, in traditional quizzes, a lot of time is required to prepare the quiz, check it, and calculate the score, which could make them less interactive. There is a need for a simple system that will enable user management, perform quizzes, score automatically, and show results in an organized manner. The QuizVerse system solves the above problem using the Java programming language.

PROJECT OBJECTIVES :

- To develop a simple and interactive quiz application using Java.
- To provide user registration and login functionality.
- To conduct multiple-choice quizzes with a time limit.
- To calculate and display the user's score automatically.
- To maintain a leaderboard based on quiz scores.
- To implement important OOP concepts such as inheritance, abstraction, and polymorphism.
- To use exception handling, collections, and multithreading in a practical application.
- To provide a basic admin panel for managing quiz information.

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FUNCTIONAL REQUIREMENTS:

1. User Registration and Login

The system shall allow students to register themselves through a username and password and allow them to log in via the same credentials. It shall also provide an independent login facility for the administrator.

2. Quiz Management

The system shall provide multiple-choice type questions containing four options each. Questions shall be kept and shuffled randomly before beginning the quiz session.

3. Timed Quiz

Each question shall contain a fixed time limit of 15 seconds. A timer shall be provided to track the time taken by the student to answer each question.

4. Answer Validation and Scoring

The system shall validate the selected answer and provide marks accordingly. Right answers will fetch points; moreover, the score will depend upon the time taken to answer the question.

5. Leaderboard

The system shall store the scores of students and arrange them in a leader board as per their total scores.

6. Admin and Error Handling

The admin shall be able to view the quiz questions available and their total number. The system shall also manage any error regarding login details and user input via exception handling.

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NON-FUNCTIONAL REQUIREMENTS :

1. Usability

The system must be user friendly and easy to use such that students can register, login, and undertake quizzes easily.

2. Performance

The system must provide quick response time to input from the user and computation of scores from the quizzes.

3. Reliability

The system must be reliable in its operations without collapsing when there are errors in data entry or login processes.

4. Security

There must be proper validation of user login details such that only registered users can have access to specific areas of the system.

5. Maintainability

The program must be simple and well organized in such a way that questions and other functionalities can be easily modified.

6. Portability

The software must be portable and capable of running in any system which has JDK installed.

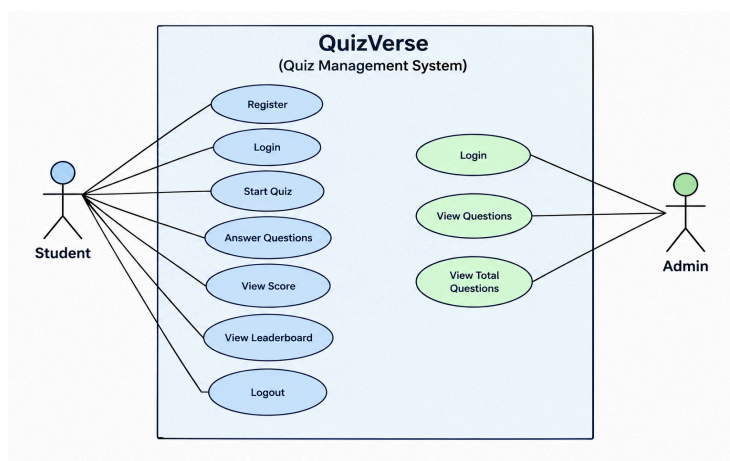
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SYSTEM ARCHITECTURE :

1. User Interface Layer – Displays the menu interface from where users can register themselves, login, take the test, and see the results.
2. User Management – Registers the users, verifies their login credentials, assigns roles to them, and logs out the users.
3. Quiz Management – Stores the quiz questions, displays them and the choices related to them, and shuffles the questions prior to the quiz.
4. Quiz Processing – Manages the timer, validates answers, calculates scores, and completes the quiz process using Java classes and multithreading.
5. Result Management – Stores the scores of the students during the execution and displays the leaderboard based on their scores.

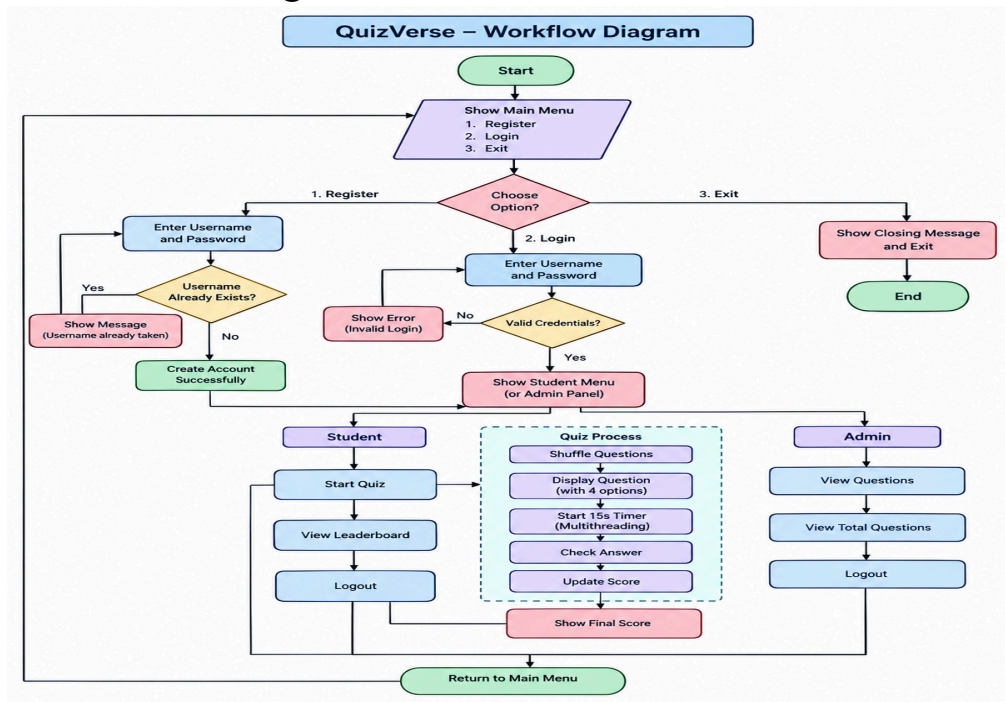
DESIGN DIAGRAM :

1. Use Case Diagram:

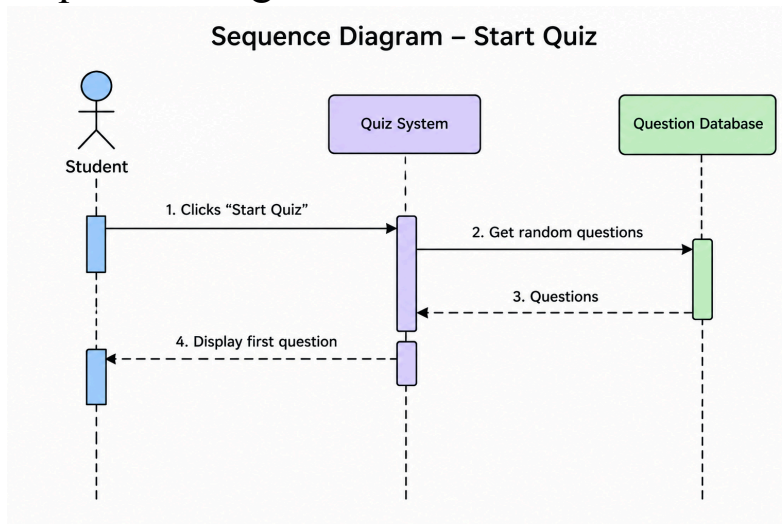


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2. Work Flow Diagram:



3. Sequence Diagram:



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DESIGN DECISIONS & RATIONALE :

1. **Object-Oriented Design:** Classes such as User, Student, Admin, and QuizQuestion are used to organize the program and apply OOP concepts like inheritance and abstraction.
2. **ArrayList:** ArrayList is used to store users and quiz questions because it allows data to be added and managed easily during program execution.
3. **Random Questions:** Questions are shuffled before each quiz to make the quiz more interesting and avoid the same question order every time
4. **Timer using Thread:** A separate thread is used for the 15-second timer to demonstrate multithreading and control the time allowed for each question.
5. **Exception Handling:** Exception handling is included to manage invalid login details and incorrect inputs without stopping the entire program.
6. **Console-Based Interface:** A simple console interface was selected because it is easy to develop, understand, and demonstrate the core Java concepts used in the project.

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IMPLEMENTATION DETAILS :

1. **Programming Language:** The QuizVerse system is implemented using Java and follows basic Object-Oriented Programming principles.
2. **User Management:** Separate classes are used for **User**, **Student**, and **Admin**. Registration and login functions are used to manage user accounts.
3. **Quiz Module:** Quiz questions are stored using **ArrayList**. Each question contains the question text, four options, and the correct answer. Questions are shuffled before the quiz starts.
4. **Timer and Scoring:** A **QuizTimer** class extends the **Thread** class to provide a 15-second timer for each question. The system checks the answer and calculates the score based on correctness and time taken.
5. **Leaderboard:** Student scores are stored during program execution and sorted to display the leaderboard according to total scores.
6. **Exception Handling:** Exceptions are used to handle invalid login details and incorrect inputs, helping the program run smoothly without unexpected termination.

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TESTING APPROACH :

1. **Registration Testing:** Checked whether a new student can register successfully and whether duplicate usernames are rejected.
2. **Login Testing:** Tested both correct and incorrect usernames and passwords to verify proper login and error messages.
3. **Quiz Testing:** Checked whether questions and options are displayed correctly and whether questions are shuffled when the quiz starts.
4. **Timer Testing:** Tested the 15-second timer to ensure that the system identifies when the time limit is exceeded.
5. **Score Testing:** Tested correct and incorrect answers to verify that scores are calculated and updated properly.
6. **Leaderboard Testing:** Checked whether student scores are displayed correctly and arranged according to their total scores.
7. **Input and Exception Testing:** Tested invalid menu choices and incorrect inputs to ensure that the program handles errors without crashing.

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SCREENSHOTS / RESULTS :

```
PS C:\Users\Admin\OneDrive\Desktop\VS CODE> & 'C:\Program Files
\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCode
DetailsInExceptionMessages' '-cp' 'C:\Users\Admin\AppData\Roamin
g\Code\User\workspaceStorage\8550401ed0d21a4d7433b81ff07c6ce3\re
dhat.java\jdt_ws\VS CODE_106367d1\bin' 'QuizVerse'
```

```
===== QUIZVERSE =====
```

1. Register
2. Login
3. Exit

Enter your choice: 1

```
----- Registration -----
```

Enter username: GokulPrasad

Enter password: gokul123

Registration successful!

```
===== QUIZVERSE =====
```

1. Register
2. Login
3. Exit

Enter your choice: 2

```
----- Login -----
```

Enter username: GokulPrasad

Enter password: gokul123

```
----- Student Menu -----
```

1. Start Quiz
2. View Leaderboard
3. Logout

Enter your choice: 1

```
===== QUIZ STARTED =====
```

You have 15 seconds for each question.

Question 1

Which method is used to start a thread?

1. run()
2. start()
3. begin()
4. execute()

Your answer: 2

Correct answer!

You got 80 points.

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Question 2

Which keyword is used for inheritance in Java?

1. implements
2. extends
3. inherit
4. super

Your answer: 2

Correct answer!

You got 65 points.

Question 3

Which OOP concept hides data?

1. Inheritance
2. Polymorphism
3. Encapsulation
4. Abstraction

Your answer: 4

Wrong answer!

Question 4

What does JVM stand for?

1. Java Virtual Machine
2. Java Variable Machine
3. Java Visual Machine
4. Java Verified Machine

Your answer: 1

Correct answer!

You got 85 points.

Question 5

Which keyword is used to create an object?

1. class
2. object
3. new
4. create

Your answer: 3

Correct answer!

You got 80 points.

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```
===== QUIZ FINISHED =====  
Score in this quiz: 310  
Your total score: 310
```

```
----- Student Menu -----  
1. Start Quiz  
2. View Leaderboard  
3. Logout  
Enter your choice: 2
```

```
===== LEADERBOARD =====  
1. GokulPrasad - 310
```

```
----- Student Menu -----  
1. Start Quiz  
2. View Leaderboard  
3. Logout  
Enter your choice: 3  
Logged out.
```

```
===== QUIZVERSE =====  
1. Register  
2. Login  
3. Exit  
Enter your choice: 3  
Thank you for using QuizVerse!  
PS C:\Users\Admin\OneDrive\Desktop\VS CODE> |
```

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CHALLENGES FACED :

- **Implementing the Timer:** Managing the 15-second timer using multithreading while allowing the user to answer the question was challenging.
- **Handling User Input:** Managing incorrect or unexpected inputs without causing the program to crash required proper exception handling.
- **Managing Scores:** Keeping track of student scores and displaying them correctly on the leaderboard required careful handling of the user data.

LEARNING AND KEY TAKEAWAYS :

- Learned how to apply **Object-Oriented Programming concepts** such as inheritance, abstraction, and polymorphism in a real project.
- Gained practical knowledge of using **ArrayList** to store and manage users and quiz questions.
- Learned how to use **multithreading** to implement a timer in Java.
- Improved understanding of **exception handling** and input validation.
- Learned how different Java concepts can be combined to develop a complete and functional application.

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FUTURE ENHANCEMENTS :

- **Database Integration:** A database can be added to permanently store user accounts, questions, scores, and quiz history.
- **Graphical User Interface:** A GUI can be developed to make QuizVerse more attractive and easier to use than the current console-based interface.

REFERENCE :

1. VITYARTHI and CSE2006 course for Java coding skills.
2. Head First Java.
3. Google for learning about Documentation.